

1 **Arteta-Ball: Modeling Arsenal’s 2024–25**
2 **Open-Play Possession as an Absorbing**
3 **Markov Chain**

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6 MATH 2210 • March 27, 2025

Abstract

This paper models Arsenal’s 2024–25 Premier League open-play possession sequences as a five-state absorbing Markov chain. We define three transient states corresponding to spatial zones on the pitch (Everywhere Else, Attacking Third, Opposition Box) and two absorbing states (Goal, Turnover). Transition probabilities are derived from season-aggregate statistics drawn from StatMuse and FBref, covering over 25,000 touches and 16,000 completed passes. We compute the transition matrix P , the initial state vector S_0 , and successive iterates S_1 , S_2 , and S_{10} to demonstrate convergence. The fundamental matrix $N = (I - Q)^{-1}$ is then used to derive absorption probabilities, yielding the key result: an open-play possession starting outside the attacking third ends in a goal with probability **6.93%** and in a turnover with probability **93.08%**, with an expected sequence length of **5.44 actions**. The model further predicts that Arsenal require approximately 14.4 such possessions to produce one goal, a figure consistent with their output of 69 league goals across 38 matches. We discuss the tactical implications for Arteta’s possession-based system and identify structural limitations of the model, including the stationarity assumption and the exclusion of set-piece sequences.

Keywords: Markov chains, sports analytics, football, absorbing states, fundamental matrix, possession modeling

1. Introduction

Football Background

Football is a game of possession. Each time a team gains the ball they begin an open-play **possession sequence**: a chain of passes, carries, and shots that ends when the ball is scored or lost. Our model covers open-play possessions only.

The pitch is divided into thirds relative to the attacking team. Once the ball crosses into the opponent's half it enters the **attacking third**; inside it sits the **opposition box**, an 18-yard rectangle in front of goal where nearly all goals are scored. Getting the ball there is harder than it sounds, and a **turnover** is what usually happens first. A **goal kick** is one common way a new possession begins, with the goalkeeper taking a free kick after the ball has gone out of play.

The Model

We model a single Arsenal open-play possession sequence as an **absorbing Markov chain**. One step is one on-ball action: a pass, a carry, or a shot. The **Markov property** says the next state depends only on the current one:

$$P(S_{t+1} = j \mid S_t = i, S_{t-1}, \dots) = P(S_{t+1} = j \mid S_t = i).$$

Transition probabilities form a 5×5 stochastic matrix P , with P_{ij} the probability of moving from state i to j and every row summing to 1. Goal (**S4**) and Turnover (**S5**) are **absorbing states**: $P_{44} = P_{55} = 1$, so every possession is guaranteed to end in one of them.

State Definitions

The five states map to zones on the pitch, chosen to match what the data can actually measure. Absorbing states are typeset in **red**. All transitions are strictly forward.

S1 — Everywhere Else. Ball with the goalkeeper, a centre-back, or a deep midfielder:

any open-play build-up phase before Arsenal enter the attacking third. Goal kicks, kickoffs, free kicks in the defensive third, and goalkeeper saves all start here.

S2 — Attacking Third. Ball has crossed into the final third. A forward or winger is building a chance.

S3 — Opposition Box. Ball inside the 18-yard box. The high self-loop ($P_{33} = 0.78$) reflects Arteta's emphasis on maintaining pressure once the team establishes itself in the box.

S4 — Goal. Possession ends with a goal or shot on target. Absorbing.

S5 — Turnover. Arsenal lose possession. Absorbing. A uniform turnover rate is applied across all zones.

2. Materials and Methods

Data Sources

All statistics are season aggregates from Arsenal’s 2024–25 Premier League campaign, drawn from StatMuse [A, D], FBref [B], and the Premier League site [C]. Season totals produce stable probability estimates. A single game would introduce too much noise.

Deriving Transition Probabilities

P_{12} is the fraction of completed passes that entered the final third: $6,479/16,360 \approx 0.396$. For P_{23} , a two-step proxy is used. Passes into the final third as a fraction of all passes gives the proportion of total touches occurring in the attacking third. Dividing box touches by this estimate gives the probability of entering the box from the attacking third:

$$P_{23} = \frac{1,364}{25,374 \times 0.396} \approx 0.136.$$

P_{34} is box touches yielding a goal: $67/1,364 \approx 0.049$. P_{24} is goals from outside the box divided by estimated non-box attacking-third touches: $2/8,683 \approx 2.3 \times 10^{-4}$. The uniform turnover rate is $4,343/25,374 \approx 0.171$ per step. Self-loops follow as $P_{ii} = 1 - \sum_{j \neq i} P_{ij}$.

Structural Assumptions

(i) *No backward transitions.* When a professional side moves forward, its shape and the forward option persist even if the ball briefly drops. Backward entries are set to zero and the self-loop absorbs temporary retreats.

(ii) $P_{13} = 0$. A direct pass from outside the attacking third into the box is near-impossible at elite level: defenders collapse before the receiver can control the ball.

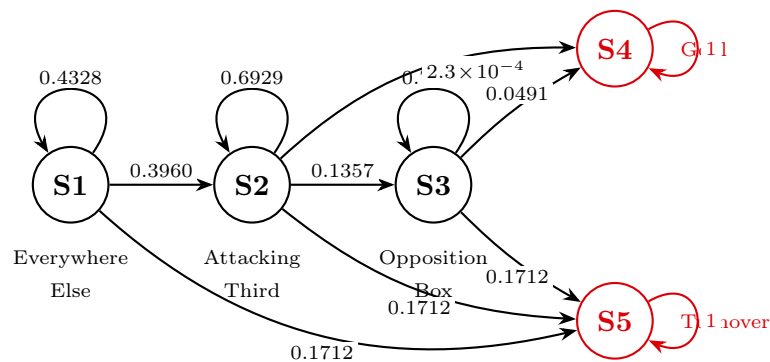
(iii) *Uniform turnover rate.* Higher risk in the final third under pressing is acknowledged, but a uniform rate keeps the model entirely data-driven.

77 The Transition Matrix

$$P = \begin{matrix} & \begin{matrix} S1 & S2 & S3 & S4 & S5 \end{matrix} \\ \begin{matrix} S1 \\ S2 \\ S3 \\ S4 \\ S5 \end{matrix} & \begin{pmatrix} 0.4328 & 0.3960 & 0 & 0 & 0.1712 \\ 0 & 0.6929 & 0.1357 & 2.3 \times 10^{-4} & 0.1712 \\ 0 & 0 & 0.7797 & 0.0491 & 0.1712 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

78 Row 2 sums to 1.00003 due to rounding in the source data, negligible for our purposes. The
 79 upper-triangular transient block encodes the no-backward-progression assumption.

80 Markov Chain Diagram



81

82 *Figure 1.* Markov chain diagram. Red nodes (**S4**, **S5**) are absorbing states; black nodes (**S1**,
 83 **S2**, **S3**) are transient.

3. Results

Initial State and Iterations

We start from **S1** with $S_0 = (1, 0, 0, 0, 0)^T$, meaning the possession begins outside the attacking third: a goal kick, kickoff, free kick in the defensive third, or goalkeeper save. Since $S_0 = e_1$, the first iterate $S_1 = P^T S_0$ is just the first row of P :

$$S_1 = (0.4328, 0.3960, 0, 0, 0.1712)^T.$$

After one action there is a **39.6 %** chance Arsenal have reached **S2**, **43.3 %** still in **S1**, and **17.1 %** immediate turnover. Reaching the box or scoring in one step is effectively zero. Computing by hand, $S_2 = S_1 P^T$:

$$S_2 \approx (0.1873, 0.4458, 0.0537, 0.0001, 0.3131)^T.$$

Probability spreads into **S2** and **S3**; turnover grows to 31 %. Via WolframAlpha:

$$\text{MatrixPower}[P, 10] \cdot \{1, 0, 0, 0, 0\}$$

$$S_{10} \approx (0.0002, 0.0385, 0.0876, 0.0459, 0.8278)^T.$$

By step 10, over 82 % of probability has been absorbed into **S5** and 4.6 % into **S4**, illustrating convergence to the steady state.

Steady State and Absorption

Because **S4** and **S5** are absorbing and every transient state can reach both in finitely many steps, a unique steady-state distribution $\bar{S}$ exists. Probability only accumulates in absorbing states, so $\bar{S}_1 = \bar{S}_2 = \bar{S}_3 = 0$ and $\bar{S}$ is supported entirely on **{S4, S5}**.

To find the split we decompose P into Q (transitions among transient states) and R (transi-

101 tions from transient to absorbing):

$$Q = \begin{pmatrix} 0.4328 & 0.3960 & 0 \\ 0 & 0.6929 & 0.1357 \\ 0 & 0 & 0.7797 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 0.1712 \\ 2.3 \times 10^{-4} & 0.1712 \\ 0.0491 & 0.1712 \end{pmatrix}$$

102 The **fundamental matrix** $N = (I - Q)^{-1}$ gives, in entry (i, j) , the expected number of steps
 103 spent in transient state j before absorption when starting from state i . Via WolframAlpha
 104 `(Inverse[IdentityMatrix[3]-Q])`:

$$N \approx \begin{pmatrix} 1.7630 & 2.2734 & 1.4004 \\ 0 & 3.2563 & 2.0058 \\ 0 & 0 & 4.5393 \end{pmatrix}.$$

105 The absorption probability matrix $B = NR$ gives the probability of eventually reaching each
 106 absorbing state:

$$B \approx \begin{pmatrix} 0.0693 & 0.9308 \\ 0.0992 & 0.9009 \\ 0.2229 & 0.7771 \end{pmatrix} \begin{matrix} \leftarrow \mathbf{S1} \\ \leftarrow \mathbf{S2} \\ \leftarrow \mathbf{S3} \end{matrix}$$

107 The steady state from $S_0 = e_1$ is therefore $\bar{S} \approx (0, 0, 0, 0.0693, 0.9308)^T$, confirmed by
 108 `MatrixPower[P,100].{1,0,0,0,0}`.

4. Discussion and Conclusion

Tactical Interpretation

Starting from **S1**, an Arsenal open-play possession ends in a goal with probability **6.93 %** and in a turnover with probability **93.08 %**. The first row of N sums to $1.7630+2.2734+1.4004 = 5.44$, so a possession starting in **S1** lasts an expected **5.44 actions** before resolving.

The diagonal entries of N are worth reading directly. A possession starting in **S1** spends on average 1.76 steps there before moving on. One starting in **S3** spends 4.54 steps in the box before resolving, which captures how tenaciously Arsenal maintain presence once they get inside the 18-yard line. This is the numerical signature of Arteta-ball.

The model puts a number on something clear to any Arsenal watcher: in open play, the main bottleneck is not finishing but getting into the box. Entry from the attacking third occurs at $P_{23} = 0.136$ per step, while Arsenal convert efficiently once inside at $P_{34} = 0.049$, consistent with 69 league goals. The self-loop $P_{33} = 0.780$ confirms that once in the box, they tend to stay.

Since each possession from **S1** scores with probability 0.0693, the model predicts Arsenal need roughly $1/0.0693 \approx 14.4$ such possessions to produce one goal. Arsenal scored 69 goals across 38 league games, averaging just under 2 per match, which is consistent with a team generating around 25–30 possessions per game and converting roughly 7 % of them: a reasonable order of magnitude for a model built entirely from aggregate passing statistics.

Limitations

- *Stationarity.* Probabilities are constant across opponents, scorelines, and match minutes. Arsenal against a low block is not the same game as Arsenal against a high press.
- *Uniform turnover rate.* One rate across all zones underestimates risk in the final third under pressing.
- *No backward transitions.* The model cannot represent a possession that regresses from the attacking third mid-sequence.

- 135 • *Open play only.* Set pieces (corners, attacking free kicks) place the ball directly in **S3**,
136 bypassing **S1** and **S2**. Arsenal are one of the league's top set-piece teams and this is a
137 meaningful gap in what the model captures.
- 138 • *Source (C).* Goals outside the box (= 2) were taken from a video compilation rather than
139 a formal statistics database.

5. Acknowledgments and References

Acknowledgments

The authors thank the MATH 2210 teaching team at Cornell University for the introduction to Markov chains that motivated this analysis.

References

- [A] StatMuse. (2025). *Arsenal 2024–25 Premier League passing statistics* [Data set]. StatMuse FC. <https://www.statmuse.com/fc/club/2024-25-arsenal-6/stats/2025?statCategory=passing>
- [B] FBref. (2025). *Arsenal 2024–25 Premier League squad statistics* [Data set]. Sports Reference LLC. <https://fbref.com/en/squads/18bb7c10/2024-2025/c9/Arsenal-Stats-Premier-League>
- [C] Premier League. (2024, September). *Every goal outside the box* [Video compilation]. Premier League Productions. <https://www.premierleague.com/en/video/4120700>
- [D] StatMuse. (2025). *Arsenal possession lost 2024–25 Premier League* [Data set]. StatMuse FC. <https://www.statmuse.com/fc/ask?q=possession+lost+by+arsenal+players+24-25&l=pl>